

ModifyAug: Data Augmentation for Virtual IMU Signal based on 3D Motion Modification Used for Real Activity Recognition

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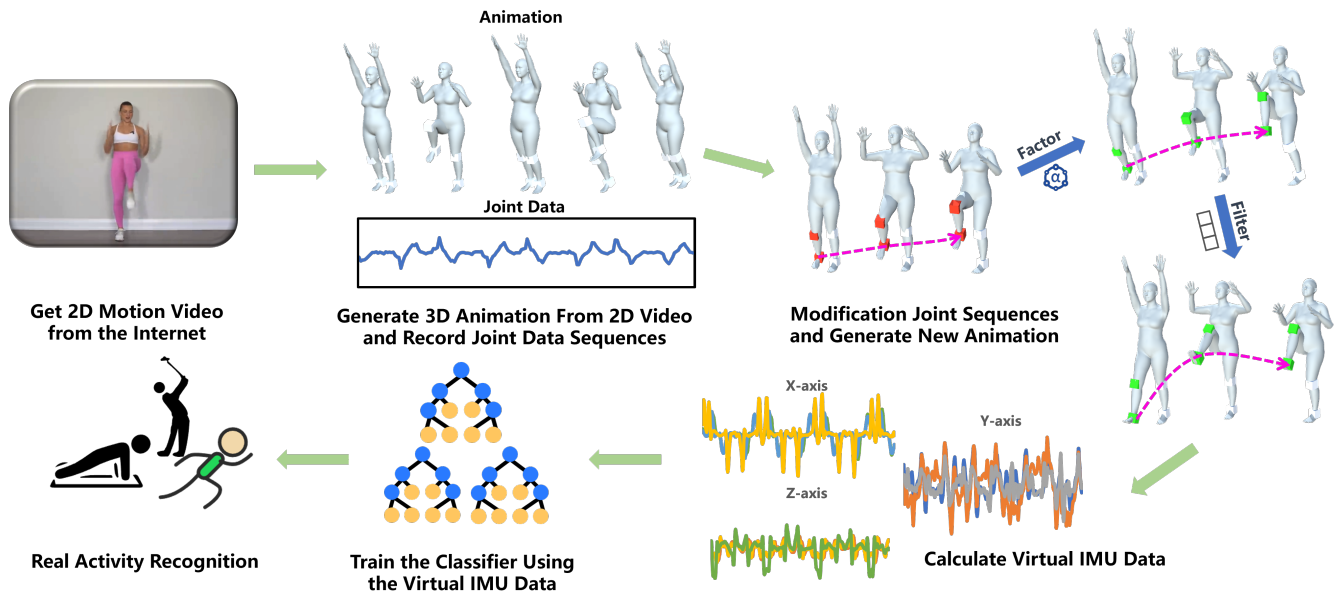


Figure 1: Overview of proposed virtual IMU data augmentation method for real activity recognition.

ABSTRACT

In wearable human activity recognition (HAR), the generation and utilization of virtual IMU data has recently gained attention. The use of virtual data can improve the robustness, effective features, and customized motion types chosen in the HAR system. However, few studies have focused on augmenting virtual IMU data to reduce the dependence on real IMU data during the machine learning model training phase. This work proposes modifying the reconstructed 3D motion by its joint data to generate larger 3D motion based on a given 3D motion sequence, thus augmenting the virtual IMU dataset. The method simulates the intra-difference of the same motion type in the real world. It aims to use fewer 3D motion inputs to generate the larger size of the virtual IMU dataset to recognize the real activity. The experiment demonstrated the feasibility of

the proposed method and provided insight into the 3D motion modification-based augmentation method.

CCS CONCEPTS

• Human-centered computing → Ubiquitous computing.

KEYWORDS

Virtual IMU Signal, 3D Avatar, Intra-Difference, Human Activity Recognition

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1 INTRODUCTION

Human Activity Recognition (HAR) has integrated into our lives, with applications from healthcare to augmented reality, and so on [1, 4, 19]. Compared to vision-based HAR methods that require the use of the cameras for recognition [15, 24], Inertial Measurement

Unit (IMU)-based HAR methods is a low-cost, convenient solution to estimate the user's posture by wearing the IMU on the user's body. The current smart devices, such as smartphones, headphones, and smartwatches, have an IMU built in, so using these devices to complete the HAR task is very convenient [13]. However, with the development of machine learning technology, IMU-based HAR technology is facing the problem of scarcity of datasets, and how to generate high-quality datasets for model training is the current research focus [8].

Facing the challenge of IMU dataset collection, the researchers explored a low-cost and flexible method to construct IMU datasets, i.e., virtual IMU data. HAR systems composed of existing datasets usually cover fixed motions, and it is very difficult for users to add custom motions. Virtual IMU data can cost-effectively augment the datasets to satisfy the user's custom motions [20]. Since different users performing the same motion will have different habits, this will lead to generating different IMU data, i.e., intra-difference, which is usually caused by various factors such as the user's age, body size, gender, and so on [18]. How to make more intra-difference of the virtual IMU data is the focus of generating high-quality virtual data, which enables the datasets to be more robust and improves the accuracy of the machine learning models.

Using a straightforward method, we implemented the virtual IMU signal augmentation based on motion video generation. We collected three motion videos to generate virtual IMU data and obtained real IMU data from seven users; the amount of data is relatively small, so the trained classifier can only classify simple motion. Experimental evaluation shows that classifiers trained on virtual IMU signals generated by our method can achieve recognition of real activities and perform better than classifiers trained using real IMU data signals. Observation of the generated IMU signals shows that the filters generate most of the intra-difference. In addition, the same filter may have different effects on different motions, perhaps causing some motions to be distorted or making some motions more realistic. However, we have yet to discuss the design of different filters for different actions to achieve better results.

In this study, we focus on how to augment the intra-difference of the data. Methods to increase the amount of data may introduce low-quality data or noise. This may affect model performance, and methods to increase the amount of data are relatively costly. Augmenting the intra-difference of the data can be achieved through data processing or data augmentation techniques, which better control the data quality and help the model focus more on learning the intrinsic features and patterns of the data. Therefore, we propose a method to convert motions from public 2D videos into 3D avatars, generate intra-difference by modifying the avatar's joint data, and then generate virtual IMU data. Finally, we collected real IMU data and the virtual IMU data generated by our method for experimental comparison, and our method achieved significant results.

2 RELATED WORK

Because of the low cost and convenience of IMU units, IMU-based HAR is currently receiving attention from researchers. Recently, IMU-based HAR has become one of the key functions in smart wearable devices; smartwatches (e.g., Apple Watch [3]) can track a user's steps and sports activity through HAR technology. Head-mounted

devices such as VR and headphones utilize HAR technology to recognize the user's head motions and body posture [7]. However, collecting IMU datasets is very difficult, and researchers have proposed some IMU data augmented methods for cost-effective and flexible construction of IMU datasets. IMUTube [9] used computer vision methods to extract the 3D motion from the 2D video, from the 3D motion generate the virtual IMU data, and then mix real data to improve the classifier's performance. Xia et al. [20] implemented a spring-joint model in the virtual environment to augment virtual IMU data. Leng et al. [10] designed a pipeline that utilizes ChatGPT [14] to generate detailed textual descriptions for HAR and uses a motion synthesis model to generate motion sequences, which are finally transformed into virtual IMU data. Due to traditional HAR datasets containing motions being relatively simple, Xiao et al. [21] introduced the AMASS dataset [12] for more fine-grained motions and enabled it to generate virtual IMU data to enhance the performance of HAR models. Li et al. [11] proposed SignRing, an IMU-based system that utilizes online videos to generate virtual IMU data and realizes automatic recognition of sign language. Xu et al. [22] proposed a physical-based data augmentation method that utilizes the physical properties of IMU sensors for data augmentation, effectively improving the generalization ability of the HAR model. Our approach utilizes a 3D avatar to achieve virtual IMU data augmentation by modifying the avatar's joint motion. The use of 3D avatar makes it easier to extend the size and activity categories of HAR datasets and makes the generated data to be interpretable.

3 MODIFY THE 3D MOTION SEQUENCES TO AUGMENT VIRTUAL IMU DATA

Fig. 1 shows our proposed data augmentation method for virtual IMU data. When a user performs a motion in the real world, the IMU signals generated from the same motion process have different amplitudes, periods, and so on. As a result of various factors of the user, making the same motion has various IMU signal types. Therefore, our method simulates the intra-difference of IMU data caused by different users for the same motion. This is done by modifying the joint data of the 3D avatar.

In the virtual environment, the motion of individual limbs for a 3D avatar can be controlled by rotation and translation. Rotation usually plays a more important role in the motion of a 3D avatar; by modifying the rotational attributes of the avatar's joints in the virtual environment, the original motion of the avatar can be changed. Therefore, we can obtain new motions by modifying the original motion sequences in the virtual environment, thus generating intra-difference and obtaining more virtual IMU data. To enable the generation of intra-difference, we first modified the amplitude of the joints using multiplication factors and later used filters to modify the joint data sequences to generate more variability in the motions. Fig. 2 shows the process of modifying the joint data to generate different motion animations by our method. We first use factors to modify the amplitude of the joint data and then use filters to further modify the joint data sequence based on this to get richer motions. The example of the original motion vs. the modified motion is shown in Fig. 3.

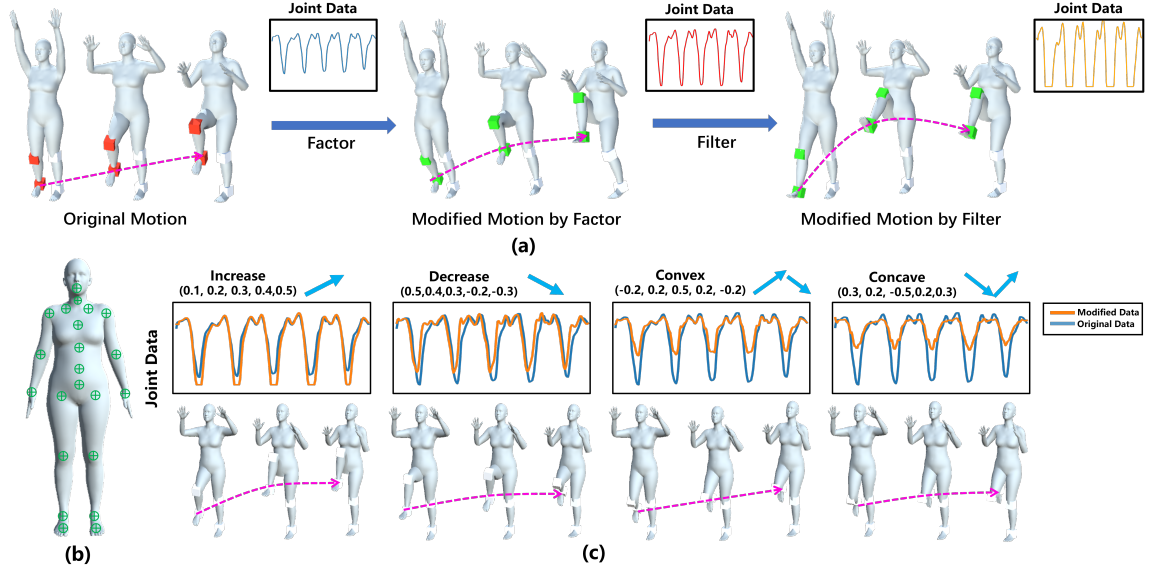


Figure 2: (a) shows the process of modifying the motion sequence by our method. (b) shows the joints included in the motion sequence, with the $\oplus$ representing the joint parts. Each of these joints can be modified using the proposed augmentation method. (c) shows the effect of filter modification on the joint data of *high knee tap*, shown for the x-axis data of the *right hip*.

3.1 Generate the Basic Motion Sequence from 2D Video

We collect the 2D video of the motion and convert it into 3D motion using *Deep-Motion* [6]. After that, the joint data during the 3D avatar's motion is loaded into the virtual environment to get the basic motion sequence of the corresponding motion. The motion sequence contains 24 joint data of the 3D avatar, which can be modified and used to generate new animations, as shown in Fig. 2.

3.2 Step1: Modify the Amplitude by Factor

Different people have different joint swings when completing the same type of motion. For example, the amplitude of knee lift is different during running. Therefore, we change the animation by modifying the amplitude of the joint motions. These different amplitudes of motion can be used for filter modification to generate richer animations. We set a multiplication factor to modify the amplitude of the 3D avatar's joint motion, and this method can be expressed by the Equation (1).

$$M_{i,axis} = \alpha O_{i,axis} \quad (1)$$

Where M is the modified joint data, α is the amplitude change factor, and O is the original joint data. To ensure that the modified joint data is normal and the 3D animation is natural, the value of α is set between 0.5 and 1.6. Fig. 4 shows the IMU data generated after the multiplication factor modification of the motion.

3.3 Step2: Modify the Motion by Filter

The multiplication factors can only affect the amplitude of the joint data. To generate more pronounced intra-difference based on

animations of different amplitudes, we utilize filters to modify the joint data sequence.

The filter is an essential concept in signal processing, and its core is the kernel. The design of the kernel will affect the function of the filter, such as high-pass filter, low-pass filter, smoothing filter, and so on. The kernel can be thought of as a function that weighs and manipulates the input signal. Different kernels can change the characteristics of the input signal, such as frequency components, amplitude, phase and so on.

When we generate a new motion sequence from the basic motion sequence, we need to change the swing amplitude and frequency of the joints, which is the same as the function of the filter. Therefore, we can consider the motion sequences as input signals and change their frequency and time domain information by designing different kernels. Different kernels can generate different animations of the same motion, thus increasing the intra-difference of the virtual IMU signal. The method of modifying the motion sequence using filters can be expressed by the Equation (2).

$$M_{i,axis} = k \otimes O_{i,axis} \quad (2)$$

Where k is the filter kernel, i and $axis$ denote the joint sequence number and the joint's axis to be modified, respectively. In practice, the filter achieves signal processing by convolving with the input signal, and the shape and window size of the kernel have an important effect on the processed signal.

To verify that the filter can implement the modification of the joint data and obtain the new 3D animations, we designed different parameter types (increase, decrease, convex, concave) for the kernel with a window size of 5. Increase means that the parameter in the window is gradually growing, decrease means that the parameter is gradually reducing, convex means that the parameter is growing

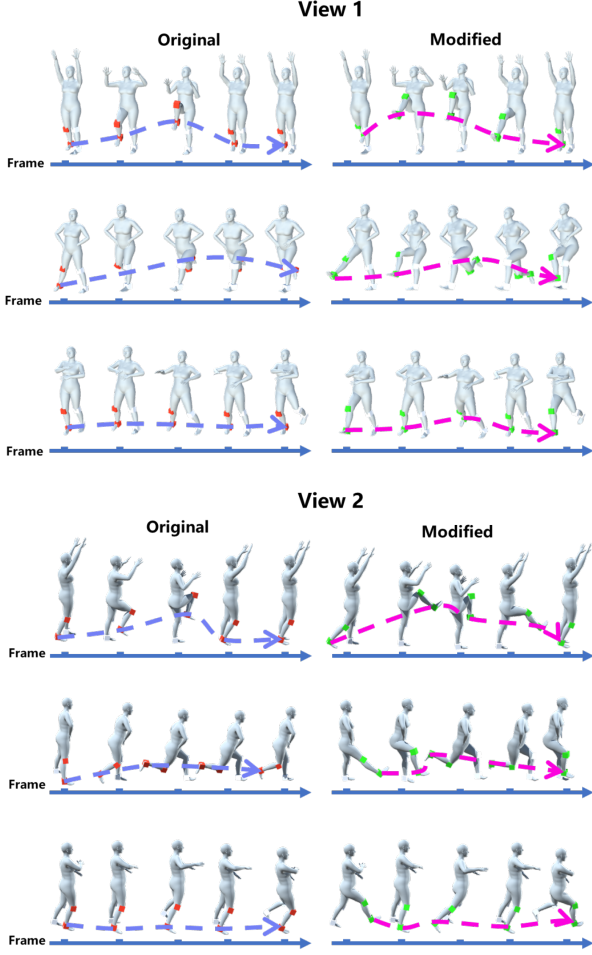


Figure 3: Original motion vs. Modified motion, where the x-axis of the avatar's right hip has been modified. The top to bottom motions are *reverse lunge*, *high knee tap* and *side to side*.

and then reducing, and concave means that the parameter is reducing and then growing. These four parameter types can compose different filters, so we use these four parameter types of filters to verify whether they can complete the joint data modification. The kernel parameters, the parameter types, and the modified joint data are shown in Fig. 2; by observing Fig. 2, due to the modification of the joint data by the filter, the modified joint data resulted in a noticeable change in the *right hip* motion of the 3D avatar. Changing the different joints allows us to get the new motion animations and new IMU data from the new animation. In addition, we artificially set the maximum motion limits of the avatar's joints in the virtual environment so that the modified motion sequences didn't show non-natural motion [5].

4 EVALUATION

Three motion videos from YouTube were selected for evaluation, each video was 25 seconds long and included the *reverse lunge*,

high knee tap, and *side to side* motions. These videos were used to generate the basic motion sequences based on the section 3.1, and our method was used to modify the basic motion sequences and generate the virtual IMU data. To evaluate the generated virtual IMU data, seven users were recruited to collect real IMU data for these motions. We installed an IMU sensor (Xsens Dot [17]) on the user's *right upper leg* and had them simulate the videos to complete the motions to get the real IMU data, each motion was recorded in 90 seconds. We used the real data to validate the effectiveness of the virtual IMU data generated by our method. In this section, we explore how to design filters for our data augmentation method and construct virtual datasets to validate the performance of our method.

4.1 Classifier Training

Virtual IMU Data: After completing the motion sequence modification, we need to calculate the virtual IMU signals based on these motion sequences. The virtual IMU data we generate is calculated based on IMUTube [9]. The virtual IMU signals used for the training model come from the *right upper leg* of the avatar, which is the same as the data acquisition part of the real data.

Classifiers and Manual Features: Due to the low dimensionality of the signal and the small amount of data, we chose to use the machine learning method based on manual features, including the Support Vector Machines (SVM) [2], Random Forests (RF) [16] and Decision Trees (DT) [23]. The manual features used are *mean*, *variance*, *standard variance*, *75th percentile*, *25th percentile*, *mean and median of the power spectrum*, *mean and median frequency of the power spectrum* [20].

Signal Adaptation from Virtual World to Real World: Due to the gap between the virtual IMU signal and the real IMU signal, to improve the robustness of the virtual IMU data, we fused the three axis acceleration data to calculate the amplitude, and then normalized to eliminate different scaling criteria on acceleration values according to [20].

4.2 Kernel Design

It can be seen from Fig. 4 that using different factors to modify the motion has less effect on the IMU signal, whereas different filters have a significant effect on the IMU signal. The window size of the kernel affects the processing range of the filter, and its parameters can be viewed as weights acting on the sample points of the input signal. Therefore, we need to experimentally evaluate how well the filter window size and parameter types affect the IMU signal to get the most suitable filter settings for motion sequence modification.

Window Size: We designed different window sizes for experimentation, each containing four different types of parameters (increasing, decreasing, convex, concave). We train the classifier on the virtual IMU signals generated by the motion modified with different window sizes and validate it on real data. The results of the experiment are shown in the Fig. 5.

From Fig. 5, it can be seen that the accuracy of the classifier trained on the virtual IMU data is highest when the window size is 3, while the accuracy of the trained classifier decreases significantly when the window size is 7. The results in Fig. 5 show that the quality of the virtual IMU data deteriorates when the window size

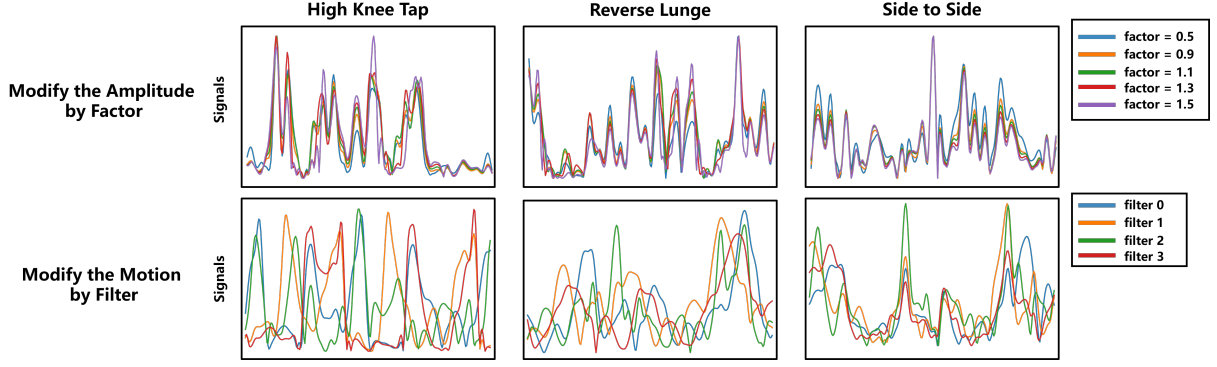


Figure 4: The virtual IMU data generated by the proposed method.

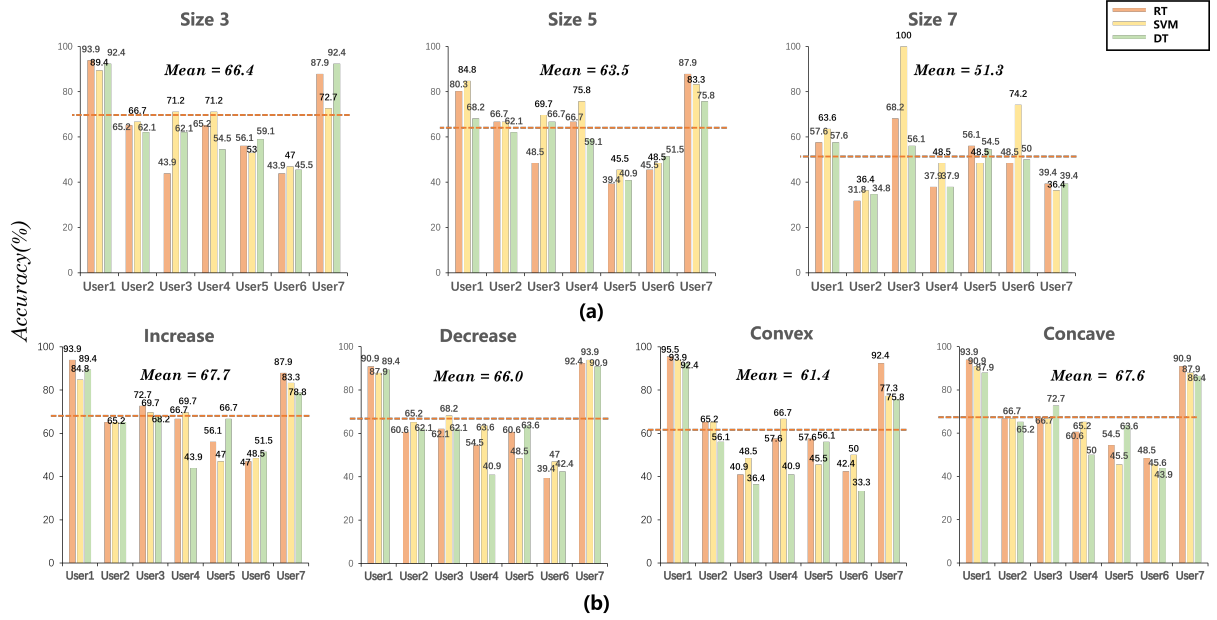


Figure 5: (a) shows the results for classifiers trained on virtual IMU data generated with different window sizes for different users. (b) shows the results for classifiers trained on virtual IMU data generated with different types of parameters for different users.

is too large. This suggests that when the window size of the filter is too large, it leads to excessive changes in the joint data sequence, generating new animations that are irrational and negative for virtual IMU data augmentation.

Types of Parameters: When designing the kernel to modify the motion sequence, in addition to the window size, the parameters of the kernel are equally important. We designed four different types of parameters for experimentation, the parameter types are increase, decrease, convex, and concave. Since the above experiments proved that the window size of 3 gives the best results, we conducted experiments on this basis and the results are shown in Fig. 5.

Observing Fig. 5, the results show that the increase and the concave types are the most effective, and convex type has the worst results. The results of the four types are not very difference, which

may be due to the differences in the results caused by the parameter settings. Therefore, all four parameter types can be used in our data augmentation methods.

4.3 Test on the Real IMU Data

We verified our data augmentation method using real IMU data from seven users as a test dataset (the total size is the 90s * 7 users * 3 motions = 1890s). We train the classifier on virtual IMU data obtained from basic motion sequences as a baseline (25s * 3 motions = 75s) and compare the classifiers trained on virtual IMU data obtained from modified motion sequences using factors and filters, respectively. The factors used in the experiments were set to 0.5, 0.7, 1.3, and 1.5 to allow for significant changes in the amplitude of the avatar. And we designed 8 different filters for experiments, set

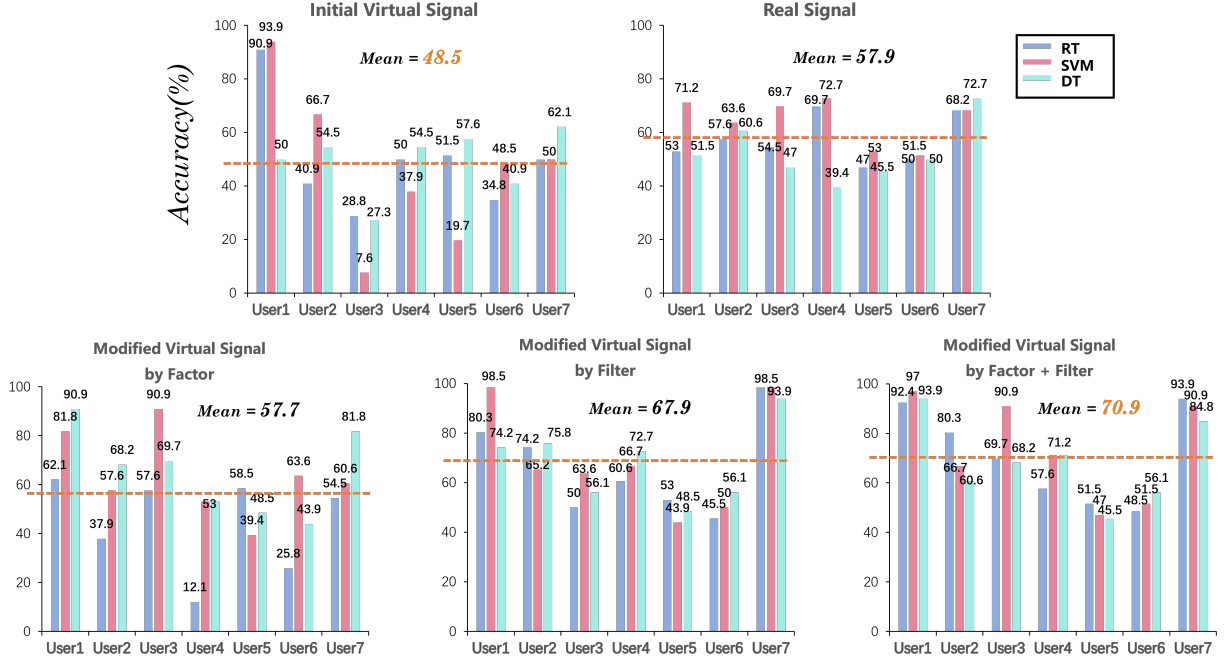


Figure 6: The results of the proposed augmentation method trained classifiers for different users.

to window size 3, with four parameter types. Finally, we compared the classifiers trained on virtual IMU data obtained using our augmentation method (using factors to modify the joint amplitude and later using filters to enrich the motions) ($25s * 4 \text{ factors} * 8 \text{ filters} * 3 \text{ motions} = 2400s$). In addition, we use the leave-one-subject-out (LOSO) method to train a classifier as the real-world baseline. All data were trained on the classifier after using the same processing methods.

The test results of the classifiers trained on different datasets are shown in Fig. 6. The classifiers trained separately using both factor and filter methods outperformed the baseline. The motions generated from the motion sequences modified by the filter method were more diverse, and resulted in the best performance. The performance of the classifier was further improved when both factors and filters were used, indicating that the intra-difference of the virtual IMU signals had become richer. We also found that the performance of the classifiers trained on the virtual dataset was better than those trained on the real dataset. This proves that the virtual IMU data generated by our data augmentation method can generate the intra-difference that works for IMU signal augmentation.

5 DISCUSSION AND CONCLUSION

In this paper, we propose a method to achieve data augmentation by modifying motion sequences to generate multiple IMU signals from a single motion sequence, resulting in a significant intra-difference. Experiments have shown that the virtual IMU data generated by our method can recognize real motions and improve the performance of the classifier.

We implemented virtual IMU signal augmentation based on motion video generation using a simple method. We collected three

motion videos to generate virtual IMU data and obtained real IMU data from seven users, the amount of data is relatively small, so the trained classifier can only classify simple motion. Experimental evaluation shows that classifiers trained on virtual IMU signals generated by our method can achieve recognition of real activities, and perform better than classifiers trained on real IMU data signals. Observation of the generated IMU signals shows that most of the intra-difference is generated by the filters. In addition, the same filter can have different effects on different motions, perhaps causing some motions to be distorted, or making some motions more realistic. Discussing the design of different filters for different actions to achieve better results could be the next step.

The method generates new animations by modifying the joint data sequences of the 3D avatars, and the 3D avatars complete the new animations to generate IMU signals. These new motion animations can generate intra-difference, but experimental results show that the method is deficient. The intra-difference generated by our method relies on the design of filters and therefore has limitations. However, it can generate a large amount of virtual IMU data, which is important for the robustness of the HAR system and the adaptation of the motion type. Investigating new methods for modifying motion sequences to generate richer and more robust motion animations is a focus of future research.

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